

GITHUB: PULL REQUESTS & CODE REVIEW

Collaborative Software Engineering — Grade 6 Session 2 Guide

Subject: Computer Science & Coding

Level: Grade 6

Provider: Planrs Academy

1. Why Teamwork Matters

In Session 1, we learned how Git acts like a digital time machine for individual coders. But what happens when you work with a team? Imagine a big school project where everyone writes their part, but no one checks the final report before turning it in. You might find spelling mistakes, or two people writing about the exact same topic! Working together on code requires coordination so everyone's work fits together seamlessly.

 **Teamwork Insight:** A great Pull Request is small and focused. It is much easier for your friends to review one specific feature than a giant pile of changes!

2. What is a Pull Request (PR)?

A **Pull Request (PR)** is a formal proposal to combine your code with the main project.

The Panchayat / School Committee Analogy

Think of a PR like proposing a new rule to a school committee or a village Panchayat. Instead of changing rules instantly, you present your proposal to the committee: *"Hey team, I've designed these changes. Can you check them before we make them permanent?"*

3. Real-World Case Study: Gully Cricket Scoring App

The 'Super Over' Feature

Imagine you and your classmates are building a **Gully Cricket scoring app**. You want to add a brand-new *Super Over* feature. Rather than pushing code directly—which might crash the app during a match—you open a Pull Request. Your teammates inspect your code, test the score calculation, and confirm it works smoothly before merging!

4. The Reviewer's Eyes: 3 Pillars of Code Review

When you are asked to review a teammate's PR, inspect the work using three key lenses:

- **1. Clarity:** Is the code easy to read and understand? Are variable names clear?
- **2. Correctness:** Does the code actually solve the problem without creating new bugs?
- **3. Kindness:** Is your feedback written in an encouraging way that helps your teammate grow?

5. The Rule of Kindness & Useful Scripts

Review the Code, Not the Coder! Never make personal comments. Focus on making the project better together.

-  **Avoid harsh comments:** "This code is wrong." or "You messed this up."

- **Use constructive scripts:**

- *"I noticed that..."* (e.g., "I noticed this loop might run endlessly if score is 0.")
- *"Could we try..."* (e.g., "Could we try renaming this variable to totalRuns?")
- *"What do you think about...?"* (e.g., "What do you think about adding a comment here?")

6. The Final Merge: Bringing It All Together

Once all team members review the PR, suggest improvements, and give their approval, it's time for the **Merge**. The merge is the digital handshake where your approved branch combines into the main branch of the project!

STUDENT WORKSHEET: COLLABORATION & CODE REVIEW

Practice conducting peer reviews, writing constructive PR comments, and planning merges!

Activity 1: The Non-Coding Peer Review — "Masala Chai Algorithm"

Step 1: Write down 3-4 steps to make a perfect cup of Masala Chai.

Step 2: Swap worksheets with your partner to review their steps. Check for missing ingredients (e.g., ginger, cardamom) or unclear boiling times!

My Chai Algorithm (Author):

Peer Review Feedback (Reviewer Partner):

Activity 2: Transforming Review Comments (Kindness Filter)

Turn these unhelpful/harsh comments into polite, constructive reviewer phrases using your Reviewer Script:

Unhelpful / Harsh Comment	Constructive & Kind Comment
"This makes no sense."	<i>e.g., "I noticed this section is a bit complex. Could we add a comment to explain it?"</i>
"Your code has bugs."	
"Redo this whole part."	

Activity 3: Pull Request Checklist & Pre-Merge Evaluation

Imagine a teammate submits a PR for a **School Sports Day Leaderboard**. Evaluate their PR:

Checklist Item	Pass / Needs Work	Reviewer Notes
Is the PR title clear? (e.g. "Add 100m Race Scores")	[] Pass [] Fix	
Is the code formatting neat and readable?	[] Pass [] Fix	
Did it break any existing scoring features?	[] Pass [] Fix	

STUDENT PRACTICE ASSESSMENT

Review & Concept Mastery Questions — Grade 6 Session 2

Instructions: Answer all questions in the space provided or on your answer sheet. No reference answers are included in this sheet.

Section A: Multiple Choice Questions

Q1. What is a Pull Request (PR) in GitHub?

- [A] A command that permanently deletes your project repository
- [B] A formal request/proposal to review code changes before merging them into the main project
- [C] A notification sent to download an updated mobile application
- [D] A automated email asking for student homework submissions

Q2. Why is it recommended to keep Pull Requests small and focused?

- [A] Because GitHub limits file sizes to 1 KB
- [B] It makes it much easier and faster for teammates to carefully review and test the specific changes
- [C] Small PRs automatically delete bugs without human review
- [D] Larger PRs cost money to upload

Q3. Which of the following best describes the 3 Pillars of Code Review?

- [A] Speed, Power, and Memory
- [B] Clarity, Correctness, and Kindness
- [C] Design, Color, and Animation
- [D] Distance, Time, and Velocity

Q4. What action takes place during the "Merge" stage?

- [A] All team members leave the GitHub project
- [B] The approved branch changes are integrated into the main project branch
- [C] The computer reboots to install system updates
- [D] The code is converted into a PDF document

Section B: Short Answer & Concept Application

Q5. How does a Pull Request resemble a village Panchayat or a school committee proposal?

Q6. Explain the rule "Review the Code, Not the Coder." Why is kindness crucial during technical peer reviews?

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Q7. Rewrite the harsh comment "This logic is completely wrong" using one of the gentle reviewer phrases learned in class.

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Section C: Real-World Scenario Problem

Q8. Scenario: Priya and Riya are working on the "Gully Cricket Scoring App". Priya creates a Pull Request titled "*Added Super Over logic*". While reviewing, Riya notices that Priya forgot to reset the wickets counter when the Super Over starts.

(a) What constructive comment should Riya write on Priya's PR using the Reviewer Script?

(b) Should Riya merge the PR immediately, or ask Priya to update the branch first? Explain why.

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